# Guanhao Wang

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Portfolio | GitHub

Computer science undergraduate focused on game development, environmental storytelling, and physical interaction.

## EDUCATION

**University of Wisconsin–Madison** June 2025 to present

Computer Science Undergraduate · Expected graduation May 2027 · GPA 3.724/4.000

**The Pennsylvania State University** August 2023 to June 2025

Undergraduate Study in Computer Science · GPA 3.94/4.00 · Transferred to UW–Madison

## GAMES AND INTERACTIVE PROJECTS

**Coal Echoes** 2026

Narrative Exploration Game · In development · Unity 6, HDRP, C#, Blender

- Designed a first person exploration game about family memories of an abandoned coal plant; built photography, album, scrapbook, and object inspection systems, enabling object discovery through the camera.
- Used an AI agent through Blender MCP to assist with parametric modeling and texture baking for industrial assets, including cableway structures and a guardhouse, then integrated them into Unity.
- Generated main menu newspaper photo materials at runtime from live HDRP Custom Pass camera captures, using HLSL halftone logic adapted from Daniel Ilett (MIT).

**Project Fireflies** 2026

Interactive Light Installation · Completed · ESP32, Arduino, Unity, UDP

- Programmed 420 LEDs across three swarms, using radar and light sensors to translate movement and ambient light into firefly disturbance and synchronization.
- Implemented two independently responsive zones, one ambient swarm, and a UDP interface for Unity synchronization commands.

**Seafloor Revival** July to August 2025

GMTK Game Jam 2025 · UI Development and Art Integration · Unity, C#

- Built all game UI and related C# behavior, integrated interface art in a six person team, and contributed to a released browser prototype of a 2.5D underwater building game.

**Mooncake** February to April 2025

Interactive Graphics · Three.js, JavaScript

- Developed a Three.js particle system and web demo with live controls for particle count and size; documented the implementation in a technical report.

## PROFESSIONAL EXPERIENCE

**IQVIA** July to August 2024

Technology Solutions Intern · Shanghai, China

- Wrote and maintained 100+ pages of user guides for a pharmaceutical WeChat Mini Program upgrade, covering two user types, four scenarios, and three device contexts.
- Performed user acceptance testing and reported 10+ defects in Jira with reproduction steps; organized 1,000+ data tags into a three level hierarchy for user filtering.

## PUBLICATION

**Human Computer Interaction in Indie Games** 2023

Sole Author

- Analyzed implicit guidance, environmental storytelling, and player attachment in FAR Lone Sails; published in Lecture Notes in Education Psychology and Public Media, vol. 18, pp. 194 to 199.

## SKILLS AND LANGUAGES

**Games and interaction** Unity, C#, HDRP, uGUI, Shader Graph, Blender, ESP32, Arduino

**Programming and tools** Python, Java, C, JavaScript, Three.js, Git, Jira

**Languages** Mandarin native, English fluent, Japanese basic